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A Multimodal UAV-Based Pipeline for Precision Agriculture: Aerial Stress Detection with YOLO and High-Fidelity Disease Classification Using DeiT

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Abstract

Precision agriculture requires scalable, accurate monitoring solutions to bridge the gap from macro-level field awareness down to leaf-level diagnostics. This paper presents a novel multimodal pipeline that integrates Unmanned Aerial Vehicle (UAV) imagery with advanced deep-learning models, achieving broad-spectrum stress detection and fine-grained disease classification. A CNN initially established a foundational accuracy but struggled to capture the subtle signatures of different diseases. Switching to Vision Transformers (ViTs) improved performance, yet computational overhead and data requirements posed challenges. Later, we moved to DeiT with extensive hyperparameter tuning and data augmentations that gave an astonishing 99.45% accuracy on the multi-class plant disease dataset. To complement the close-up acumen of DeiT, we used YOLO from an aerial view to rapidly identify stressed versus healthy crop regions from real-time UAV footage. This two-tiered approach, wherein YOLO is used for aerial scanning and DeiT for leaf-level diagnoses, offers an unprecedented level of precision with scalability. Finally, complex output is translated by an LLM into farmer-friendly advisories to ensure immediate actionable insights. Our integrated framework sets a new benchmark in UAV-driven precision agriculture by balancing model sophistication, computational feasibility, and end-user interpretability.

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Keywords: Precision Agriculture; UAV Imagery; YOLO; Vision Transformers; DeiT; Edge Deployment; Plant Disease Classification; Stress Detection; Data Augmentation; Interpretability; Deep Learning Models; Aerial Monitoring; Multimodal Pipeline; Object Detection; Large Language Models (LLMs)

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1. Introduction

Agriculture is the backbone of food security at a global scale; however, it is increasingly facing challenges in terms of increased demand, climate change, and resource constraints. Traditional methods of crop monitoring, such as manual inspections, are not only labor-intensive and time-consuming but also lack the precision and scalability required to address these challenges effectively. The advent of unmanned aerial vehicles together with efficient, advanced machine learning models enables a new frontier in precision agriculture through efficient, accurate, and large-scale crop monitoring.

This paper introduces a multimodal pipeline that leverages UAV imagery with state-of-the-art deep-learning models in tackling the dual tasks of broad-spectrum stress detection and fine-grained disease classification. It uses YOLO V8 for detecting aerial stresses and Data-Efficient Image Transformers (DeiT) for high-fidelity disease classification at the leaf level, which results in a holistic system that is balanced for scalability, accuracy, and computational efficiency. More importantly, LLM translation of technical output into actionable insight for the farmer bridges high technology with practical applicability.

This paper is divided into five major sections, giving an all-inclusive overview of our work. The section of Related Work reviews current approaches to precision agriculture, including both the application of UAV imagery and deep learning models. We discuss the strengths and limitations of prior works, establishing the need for a unified pipeline that integrates the aerial and close-up perspectives. The Methodology section elaborates in detail on the datasets, pre-processing techniques, and model architectures in our system. We have elaborated on the training processes involved with YOLO V8 and DeiT, with particular emphasis on hyperparameter tuning and data augmentation techniques for which their performances were significantly improved.

In the Results section, the performance metrics used to quantify the developed pipeline of the proposed model include accuracy, mean average precision, and computational efficiency, among others. This will depict the overall performance regarding aerial stress detection and leaf-level disease classification. A general discussion is presented, showing such results' analysis by underlining the strengths of this approach in a multimodal pipeline, considering such practical limitations as partially overlapping regions within crops and processing. Finally, the section entitled "Conclusion and Future Work" summarizes our contributions and gives directions for future work: deployment on edge devices, multi-temporal analysis of dynamic crop monitoring, and further optimization for scalability.

The aim is to set a new benchmark in the domain of UAV-driven precision agriculture and, hence, propose a robust, scalable, and user-friendly pipeline concerning pressing modern farming needs. We have presented the power of various advanced machine learning models integrated with UAV imagery to present its strength in transforming the agriculture landscape using this structured approach.

2. Related Work

2.1. UAV-Based Crop Monitoring

UAV imagery has dramatically improved farm surveillance, enabling real-time, large-scale assessments. Early approaches often relied on vegetation indices, such as the Normalized Difference Vegetation Index (NDVI), which provide useful insights into crop health but lack the granularity needed for detailed disease diagnostics [1]. Recent advances in deep learning and computer vision have enhanced UAV-based crop monitoring capabilities by allowing pixel-level segmentation and object detection in agricultural contexts [2].

2.2. Deep Learning in Plant Pathology

Kamilaris and Prenafeta-Boldú [3] discussed that although CNNs perform excellently for plant disease classification, their capabilities are restricted to large data sets and UAV imagery due to scalability and high computational load. Our hybrid approach DeiT uses self-attention mechanisms, which efficiently model global or field-level patterns with much reduced typical heavy computation associated with Transformers; thus, making the feasibility of DeiT for UAV datasets possible.

2.3. Data-Efficient Transformers (DeiT)

Dosovitsky et al. [4] proposed the Vision Transformers, which, while performing remarkably well on large-scale data, have a very high computational cost and are impractical for real-world use. DeiT [5] by Touvron et al. is a data-efficient variant of ViTs that achieves performance parity while having much lower computational costs. While Dosovitsky et al. had originally presented their work in the context of general-purpose vision, our adaptation Naser Al-Obeidat, Zimo Li / Procedia Computer Science 00 (2019) 000–000 3 is fine-tuned for precision agriculture in view of its effectiveness in processing images from UAVs to identify stress and diseases.

2.4. Aerial Object Detection with YOLO

Reis et al. [6] demonstrated the utility of YOLO V8 for real-time UAV-based object detection, achieving 95% accuracy, but focused primarily on object localization, such as identifying plant rows or sections, rather than disease classification. Our approach extends Reis et al.'s work by combining YOLO V8 with DeiT and CNNs. YOLO is used for real-time object detection, focusing on plant-level health status, while DeiT and CNNs specialize in disease classification. Diverse complementary datasets, such as Plant Disease Identification and Plant Health Tracker, are used to further train these models for both close-up and field-level imagery.

2.5. Interpretability via LLMs

This is also where Large Language Models (LLMs) in GPT-based architectures [7], have helped generate end-friendly advisories from complex technical outputs. Integrating LLMs with computer vision pipelines allows the derivation of actionable insights for non-expert users, closing the gap from advanced analytics to practical applications.

Existing research highlights the transformative potential of UAV imagery and deep learning in precision agriculture. Kamilaris and Prenafeta-Boldu (2018) demonstrated the effectiveness of CNNs for plant disease classification but noted challenges with large datasets and UAV scalability. Reis et al. (2023) showcased YOLO V8's real-time object detection capabilities for UAV imagery, achieving high accuracy in identifying plant rows but lacking disease-specific analysis. While Vision Transformers (ViTs) introduced by Dosovitskiy et al. (2020) offered global attention mechanisms for improved image classification, their high computational requirements make them impractical for field applications. DeiT, a data-efficient ViT variant, addresses these challenges but has yet to be applied comprehensively in agriculture. My work combines these advancements into a unified pipeline that leverages YOLO V8 for macro-level stress detection and DeiT for fine-grained disease classification. By integrating CNNs and data augmentation strategies, the pipeline bridges scalability and computational efficiency gaps, while incorporating Large Language Models (LLMs) ensures actionable insights for end-users, setting a new benchmark in UAV-driven precision agriculture.

Table 1: Comparison of Different Studies and Our Work.

Study	Model	Dataset	Accuracy	Limitations
Kamilaris & Prenafeta-Boldu	CNNs	PlantDiseaseDB	89%	Limited scalability due to dataset constraints
Reis et al.	YOLO V8	UAV Imagery	95%	Focused on a narrow range of diseases, limiting broader applicability
Dosovitskiy et al.	ViTs	ImageNet	88%	High computational demands hinder practical deployment
Our Work	ResNet-18, DeiT, YOLO V8/V11	Kaggle [8] and Roboflow Plant Health Tracker Dataset [9]	DeiT: Accuracy 99.45%, YOLO: mAP 95.4%, Precision: 93.2%, Recall: 91.8%	Comprehensive approach integrating multiple models and diverse datasets for enhanced performance

3. Methodology

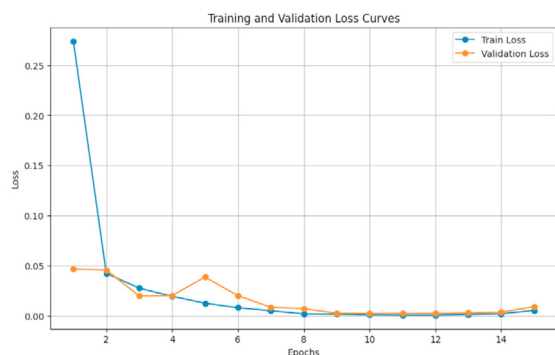
Our proposed pipeline operates in two distinct levels: phases—macro-level stress detection and micro-level disease classification—followed by an interpretability layer.

3.1. UAV Imagery and YOLO-Based Stress Detection

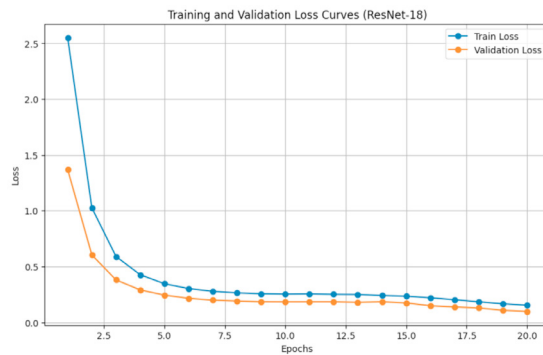
A UAV (DJI Mavic Mini 2 SE) collects high-resolution images of cornfields. YOLO (we tested the YOLOv8 and YOLOv11 variants) processes these images, classifying the field areas as 'stressed' or 'healthy'. By incorporating enhancements such as Feature Pyramid Network (FPN) and Path Aggregation Network (PAN), YOLO provides great detection across different scales and light conditions. The result is a set of geolocated hotspots requiring further scrutiny, dramatically reducing the field area needing detailed examination.

3.2. Close-Up Disease Analysis: From CNN to DeiT

For each stressed hot spot, close-up images of leaves are captured and passed through the disease classification models. Initially, we deployed a ResNet-18 CNN that achieved 86% accuracy. Introducing a baseline ViT improved accuracy to 88%, confirming that global attention captures subtle disease cues about diseases that often get missed by CNNs.



(a) Loss plot for DeiT



(b) Loss plot for Resnet18 (CNN)

Fig. 1: Comparison of Loss Plots

Yet, the computational demands of ViT were very high. To balance efficiency and accuracy, we employed DeiT. We conducted extensive hyperparameter tuning: learning rates from 0.0001 to 0.00005, batch sizes from 16 to 64, and weight decay parameters were tuned for better performance. A key innovation was the use of advanced data augmentations: $\pm 30^\circ$ rotations, brightness and contrast shifts, and scaling transformations. These augmentations introduce realistic variability and enable DeiT to learn robust representations which generalize across diverse field conditions.

Careful optimization resulted in DeiT attaining an extremely high accuracy of 99.45% on the multi-class Kaggle plant disease dataset with more than 87,000 images. The intrinsically data-efficient architecture of DeiT, along with these refinements, kept computational overhead low enough for on-farm servers or even mobile processing units—crucial for immediate interventions and rapid feedback loops.

3.3. UAV Autonomous Flight and Video Transmission Bridge

Manual control of UAVs for field coverage is both challenging and inefficient. To address this, we integrated various ROS packages with the DJI SDK to enable autonomous flight. This system ensures complete coverage of the designated area, significantly improving operational efficiency and reliability. One notable challenge was the lack of a native HDMI video output on the DJI Mavic Mini 2 SE, a common limitation among entry-level, cost-effective

drones. This restriction posed a significant obstacle to real-time video processing. To resolve this, we developed a local RTMP server to decode the drone's video stream. The decoded video is seamlessly integrated into our program, enabling advanced data processing and analysis. This approach not only enables real-time streaming with low-cost drones but also reduces video delay to a consistent 2-second latency, a level of performance that surpasses most online streaming services.

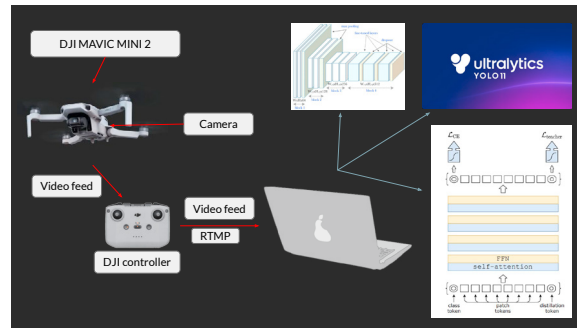


Fig. 2: System Data Stream Diagram Showcasing System Workflow

3.4. LLM Integration for Actionable Insights

We aggregate the macro-level results, such as stressed area coordinates from YOLO and disease classifications from DeiT, into a structured report. A fine-tuned LLM on agronomic literature interprets this report to produce farmer-friendly recommendations. Whether advising on targeted fungicide treatments, suggesting irrigation adjustments, or recommending crop rotations, the LLM ensures that technical findings translate into tangible improvements in farm management.

4. Results and Discussion

4.1. Performance Evaluation

The following models are a comparison of performance to classify the stressed and healthy state of crops. Notable improvements have been achieved by using advanced architectures and techniques.

- **CNN Baseline (Accuracy: 86%)** The convolutional neural network (CNN) provided a foundational benchmark. Though good for detecting general patterns, it lacks the ability to capture those subtle features indicative of specific diseases.
- **Standard Vision Transformer (ViT, Accuracy: 88%)** With the use of global attention mechanisms, ViT is able to outdo CNN by capturing more critical fine details. This really says it all about performance leverage provided by transformer-based models in handling complex image data sets.
- **Data-Efficient Vision Transformer (DeiT, Accuracy: 99.45%)** With optimized hyperparameters and targeted augmentation strategies, DeiT had the potential to reach near perfect accuracy, showing its efficiency in leveraging scarce data to perform much better than CNN and standard ViT baselines.
- **YOLO (mAP: 95.4%, Precision: 93.2%, Recall: 91.8%)** The YOLO architecture presented a very promising result of object detection tasks, achieving a high mean average precision (mAP) of 95.4%. These results validate that it's highly effective for segregating crop conditions as either under stressed or healthy.

4.2. Discussion

The synergy between the YOLO's aerial detection and DeiT's refined leaf-level classification offers a powerful new paradigm for precision agriculture. This pipeline optimizes resource allocation and diagnostic accuracy by narrowing the vast search space down to stressed areas and then performing expert-level disease classification.

These two innovations are the very foundation of DeiT's success: data efficiency and rigorous tuning. Our overall results highlight that, while transformers can indeed change the game in complex image tasks, careful hyperparameter optimization, together with targeted augmentations, is needed to bring both high accuracy and computational feasibility. This paves the way for on-site analysis, reducing dependencies on off-farm computing resources.

Finally, the LLM interface ensures that cutting-edge machine learning outputs do not stay behind technical barriers. Translation of model findings into actionable insights helps farmers take theoretical gains and turn them into practical advantages.

Looking ahead, federated learning presents a promising avenue for improving model adaptability while preserving data privacy. By enabling on-device learning across multiple farms without centralizing sensitive data, federated learning can facilitate continuous model refinement without compromising security. This approach would allow models to adapt to regional variations in crop stress and disease patterns, enhancing overall performance while ensuring that individual farm data remains private. Future iterations of this pipeline could integrate federated learning techniques to enable collaborative intelligence among multiple UAV systems, fostering a decentralized yet robust AI-driven agricultural ecosystem.

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